Machine learning and India VIX: can a model time crash insurance better than VIX?

FDRM course project, Group 8, IIT Jodhpur · github.com/Subham1709/VIX_ML

Question

Our main project protected a Nifty 50 portfolio with a 3-month put, 5% below the market, bought only when India VIX was below 13 and sold into volatility spikes. It returned 13.9% a year against 11.1% unhedged, with a worst fall of 22% instead of 38%. This extension asks whether machine learning (gradient boosting) or deep learning (an LSTM) can time that insurance better. We tried two ideas: buy when a model expects next month's volatility to be at least what VIX charges (*insurance looks cheap*), and buy when a model's chance of a 7% fall within three months is above its long-run rate (*fall risk above normal*).

Data and method

Nifty 50 daily prices from September 2007 and India VIX from January 2020 to 13 April 2026 (the main project's files). Each day the models see 15 Nifty features known at the close (realised volatility over 5 to 126 days, high-low range volatility, downside volatility, 1-week to 3-month returns, distance from the 1-year high and moving averages); VIX versions add 5 VIX features. The LSTM reads the last 63 days.

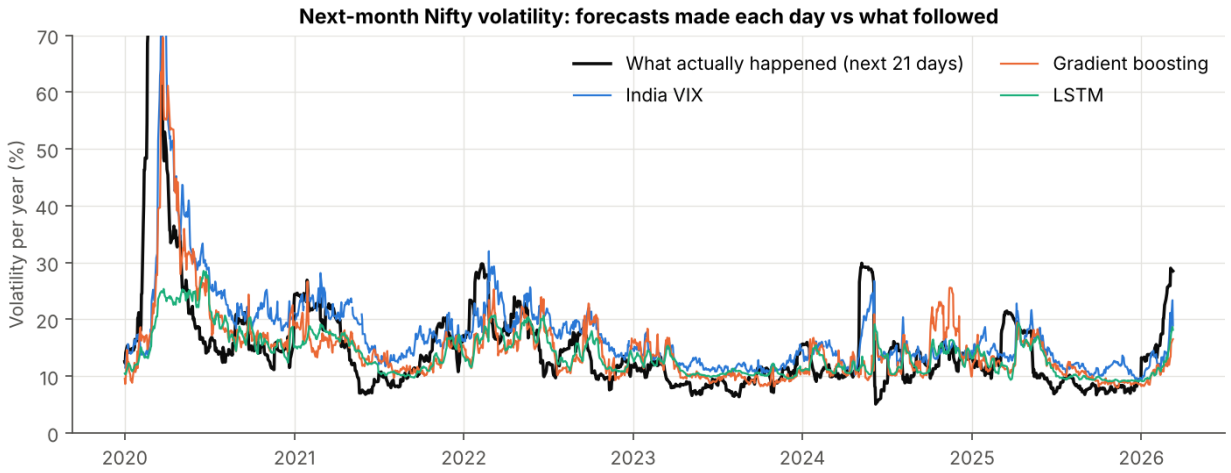
No lookahead. Forecasts are walk-forward: on the first trading day of each month every model is retrained on earlier data only (the LSTM every three months), and a day enters training only after its answer is known (21 days later for volatility, 63 for falls). The test runs from January 2020, after twelve years of training data. Model settings and the two buying thresholds were fixed before the backtest. Option pricing and trading rules are those of the main project; our engine reproduces its original-rule results exactly.

Forecasting volatility

Forecast of next-month volatility (Jan 2022 to Mar 2026)	Average error (vol pts)	Bias (vol pts)	Improvement on VIX	Significance (t)
India VIX	3.75	+2.12		
Last month's volatility	3.95	-0.15	-31%	-1.00
HAR regression	3.45	+0.17	+1%	0.05
HAR + VIX regression	3.33	-0.22	+15%	1.31
Gradient boosting (Nifty only)	3.81	+0.05	-8%	-0.51
Gradient boosting (Nifty + VIX)	3.76	-0.22	-7%	-0.51
LSTM (Nifty only)	3.31	-0.18	+10%	0.68

Improvement on VIX: share of VIX's squared log error removed. Significance: Diebold-Mariano t-statistic with Newey-West errors (about 2 is needed).

The LSTM's error was 3.3 points against 3.8 for VIX, largely because VIX is on average 2.1 points too high (the volatility risk premium). A simple regression mixing recent volatility with VIX did as well; gradient boosting did not beat VIX. None of the improvements is statistically significant.



Predicting falls

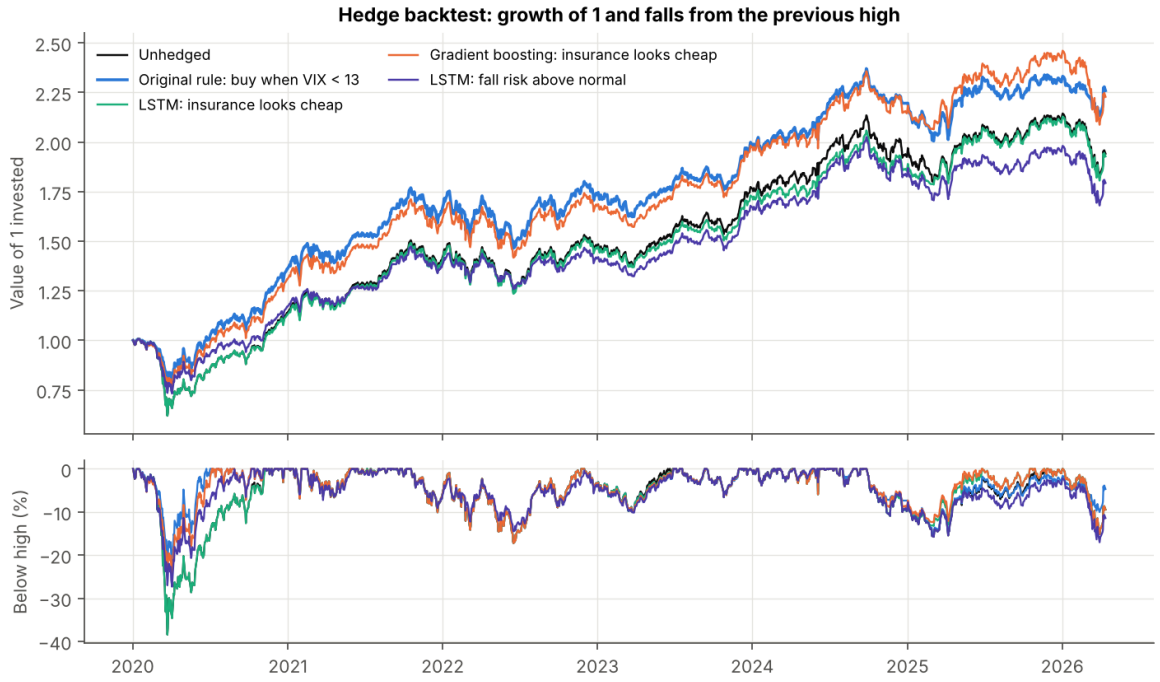
Chance of a 7% fall within 3 months (Jan 2022 to Mar 2026)	AUC (0.5 = guessing)	Brier score
Base rate	0.48	0.165
Logistic on last month's vol	0.62	0.145
Logistic on India VIX	0.23	0.160
Logistic (all Nifty features)	0.51	0.153
Gradient boosting (Nifty only)	0.56	0.171
LSTM (Nifty only)	0.60	0.151

No model ranks the days before a fall much better than chance. The logistic model on VIX, trained on 2020-21, learned that high VIX precedes falls (true in the COVID crash) and was then wrong: after 2021 most falls began in calm markets, exactly when the original rule buys insurance.

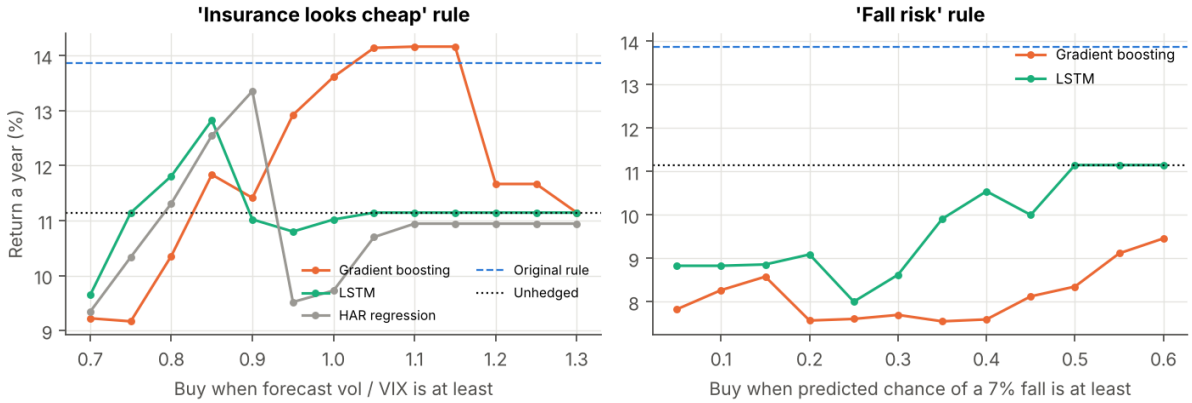
Timing the hedge

Strategy (Jan 2020 to Apr 2026)	Return a year	Worst fall	Sharpe	Insurance cost a year	Puts	Return from Jul 2020
Unhedged	11.1%	-38.4%	0.25	+0.0%	0	15.4%
Original rule: buy when VIX < 13	13.9%	-22.0%	0.48	-2.2%	22	14.9%
Buy every month	8.8%	-25.5%	0.17	+2.2%	73	11.4%
Gradient boosting: insurance looks cheap	13.6%	-24.9%	0.44	-2.0%	9	15.3%
LSTM: insurance looks cheap	11.0%	-38.4%	0.25	+0.1%	5	15.2%
Gradient boosting: fall risk above normal	8.1%	-38.4%	0.10	+2.8%	33	13.2%
LSTM: fall risk above normal	9.8%	-27.3%	0.22	+1.4%	34	12.1%
VIX < 13 or LSTM insurance cheap	14.2%	-22.0%	0.50	-2.4%	25	15.2%

All strategies use the same put and sell half at VIX 22, the rest when VIX falls 20% from its peak. Negative cost: the puts earned more than they cost.



The best pre-set ML rule, gradient boosting *insurance looks cheap*, returned 13.6% against 13.9% for the original rule, with a deeper worst fall (-25% vs -22%). The LSTM version did not buy before the COVID crash and ended level with no hedge (11.0%). Rules based on fall risk did worse than no hedge. Combining the original rule with the LSTM signal added 0.3 points a year, too little to trust given the fragility below.



Moving a threshold by 0.05 changes returns by up to 4 points, because one put more or fewer in early 2020 decides the result. From July 2020 the original and both 'cheap insurance' rules are within about half a point a year of holding Nifty unhedged.

Conclusion

Machine learning produced slightly better volatility forecasts than VIX, but no reliable improvement in predicting falls or in timing the insurance. The simple rule of buying puts when VIX says they are cheap remains the best performer, which supports the main project's conclusion: what matters is the price of insurance, not a forecast of the next crash.

Limitations

Option prices are modelled from VIX rather than traded prices. VIX starts in 2020, so VIX-based models have one to four years of training data. Six years contain one real crash, and overlapping 3-month windows hold far fewer independent observations than the 1,500 test days suggest. Trades are at the close on the day of the forecast, as in the main project.